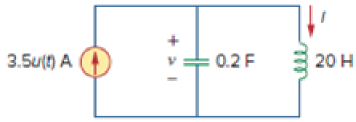
Chapter 8, Problem 8PP

Problem

Find $i(t)$ and $v(t)$ for $t > 0$ in the circuit of Fig. 8.24.

Figure 8.24:



Step-by-step solution

Step 1 of 8

Refer to Figure 8.24 from the textbook.

Draw the modified circuit for $t<0$.

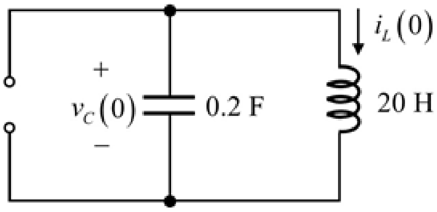


Figure 1

For $t<0$, the step function, $u(t)$ is zero that is the current source $3.5u(t)$ is open circuit circuited. Hence the circuit becomes source free circuit.

Therefore, the initial values are,

$$i_L(0)=0 \text{ A}$$
$$v_C(0)=0 \text{ V}$$

Step 2 of 8

At $t=\infty$, as steady state is reached, capacitor is open circuited and inductor is short circuited.

Draw the modified circuit at $t=\infty$.

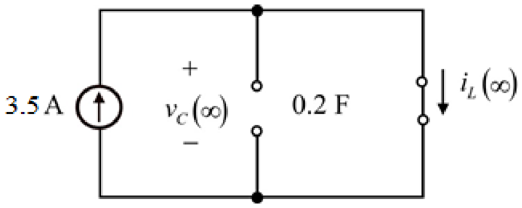


Figure 2

Calculate the steady state value of the current.

$$\begin{aligned} i_{ss}(t) &= i(\infty) \\ &= 3.5 \text{ A} \end{aligned}$$

Step 3 of 8

Draw the modified circuit for $t \geq 0$.

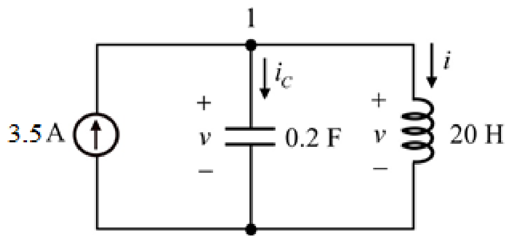


Figure 3

Step 4 of 8

For $t \geq 0$, the step function, $u(t)$ is one that is the current source $3.5u(t)$ is 3.5 A and the voltage across the inductor and capacitor is same because of the parallel circuit.

$$v = L\frac{di}{dt}$$

Apply Kirchhoff's current law at the node 1.

$$i_C + i = 3.5$$
$$C\frac{dv}{dt} + i = 3.5$$

Substitute $L\frac{di}{dt}$ for v.

$$C\frac{d}{dt}\left(L\frac{di}{dt}\right) + i = 3.5$$
$$LC\frac{d^2i}{dt^2} + i = 3.5$$
$$\frac{d^2i}{dt^2} + \frac{i}{LC} = \frac{3.5}{LC}$$

Step 5 of 8

Substitute 0.2 F for C and 20 H for L.

$$\frac{d^2i}{dt^2} + \frac{i}{(0.2)(20)} = \frac{3.5}{(0.2)(20)}$$
$$\frac{d^2i}{dt^2} + \frac{i}{4} = \frac{3.5}{4}$$
$$\frac{d^2i}{dt^2} + 0.25i = 0.875$$

The associated polynomial is,

$$s^2 + 0.25 = 0$$
$$s = +j0.5,\ -j0.5$$

Step 6 of 8

Calculate the cut-off frequency, ω_0.

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{LC}}\\ &= \frac{1}{\sqrt{(0.2)(20)}}\\ &= \frac{1}{\sqrt{4}}\\ &= 0.5\end{aligned}$$

Therefore, the expression for the current, $i(t)$ is,

$$\begin{aligned}i(t) &= i_{ss}(t) + i_t(t)\\ &= i_{ss}(t) + \left[A_1\cos(\omega_d t) + A_2\sin(\omega_d t)\right]\\ &= 3.5 + \left[A_1\cos(0.5t) + A_2\sin(0.5t)\right]\end{aligned}$$

Substitute 0 for t.

$$\begin{aligned}&i(0) = 3.5 + \left[A_1\cos 0.5(0) + A_2\sin 0.5(0)\right]\\ &0 = 3.5 + A_1\\ &A_1 = -3.5\end{aligned}$$

Step 7 of 8

Differentiate the function $i(t)$ with respect to t.

$$\begin{aligned}\frac{di(t)}{dt} &= \frac{d}{dt}\left[3.5 + \left(A_1\cos(0.5t) + A_2\sin(0.5t)\right)\right]\\ &= -A_1\sin(0.5t) + A_2\cos(0.5t)\end{aligned}$$

Substitute 0 for t.

$$\begin{aligned}&\frac{di(0)}{dt} = -A_1\sin 0.5(0) + A_2\cos 0.5(0)\\ &0 = A_2\\ &A_2 = 0\end{aligned}$$

Substitute 0 for A_2 and -3.5 for A_1 in $i(t)$.

$$\begin{aligned}i(t) &= 3.5 + \left[(-3.5)\cos(0.5t) + (0)\sin(0.5t)\right]\\ &= 3.5 - 3.5\cos(0.5t)\\ &= 3.5\left[1 - \cos(0.5t)\right]\end{aligned}$$

Therefore, the expression for the $i(t)$ for $t > 0$ is $\boxed{3.5\left[1-\cos(0.5t)\right]\text{ A}}$.

Step 8 of 8

Calculate the voltage, $v(t)$.

$$v(t) = L\frac{di(t)}{dt}$$

Substitute $3.5(1-\cos 0.5t)$ for $i(t)$ and 20 H for L.

$$\begin{aligned} v(t) &= 20\frac{d}{dt}\left[3.5\left(1-\cos(0.5t)\right)\right] \\ &= 70\frac{d}{dt}\left[1-\cos(0.5t)\right] \\ &= 70\left[0-0.5\left(-\sin(0.5t)\right)\right] \\ &= 35\sin(0.5t) \end{aligned}$$

Therefore, the expression for the voltage, $v(t)$ is $\boxed{35\sin(0.5t)\text{ A}}$.